

Origin and fate of dietary nanoparticles and microparticles in the gastrointestinal tract.

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Humans have evolved with oral exposure to dietary microparticles and nanoparticles as a normal occurrence but the ever-growing exploitation of nanotechnology is likely to increase exposure further, both qualitatively and quantitatively. Moreover, unlike the situation with respirable particles, relatively little is known about gastrointestinal intake and handling of nanoparticles. With a long term interest in gut exposure and responses to dietary microparticles, our group is now applying its expertise to nanoparticles in the gastrointestinal tract. Here we aim to address (i) the current challenges associated with the characterisation of particle-host or particle-cell interactions, (ii) the origin and mechanisms of uptake of particles in the gastrointestinal tract, especially via the Peyer's patch and (iii) potential cellular effects of nanoparticles in the generation of reactive oxygen species and inflammasome activation, or microparticles in their adjuvant activity in pro-inflammatory signalling and immune responsiveness. Copyright 2010 Elsevier Ltd. All rights reserved.